

How Air Quality Drives Hospital Visits Across Kansas

A Random Forest Regression Approach Linking PM2.5 to
Number of Hospital Visits

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Motivation: Does Air Pollution Drive Healthcare Visits?

- Kansas communities face varying levels of fine particulate matter (PM_{2.5})
- **Key Question:** Does higher pollution exposure lead to more healthcare utilization?
- **Significant Challenges:**
 - Geographic mismatch between pollution data and visit data (census block)
 - Confounding demographic factors (race, smoking, digital access)
 - Non-linear relationships between pollution and health outcomes
- **Goal:** Quantify the relationship between PM_{2.5} and total healthcare visits while controlling for sociodemographic factors

Pollution Data (df1):

- Kansas PM2.5 estimates 2016–2020
- Variables: `ds_pm_pred`, `ds_pm_stdd`

Healthcare Data (df2):

- `patients`, `encounters`, `social_determinants`
- Visit counts by census block
- Transformed: `TotalVisits`, `UniquePatients`, `UniqueJourneys`
- Outliers $> 30,000$ visits removed

External Data - Retrieved from [Data.CDC.gov](https://data.cdc.gov)

Geographic Evidence: Pollution Maps Visits

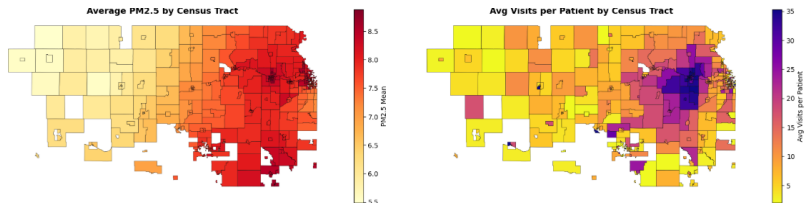


Figure 1: Left —Average PM2.5 by census tract. Right —Avg visits per patient. Eastern tracts show high values on both maps.

Key insight: The eastern concentration of both high PM2.5 and high Hospital Visits perfectly aligns with our hypothesis.

Statistical Evidence: PM2.5 and Hospital Visits

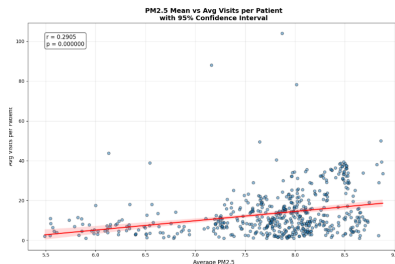


Figure 2: PM2.5 Mean vs Avg Visits per Patient with 95% Confidence Interval

Correlation Test:

- $r = 0.2905$
- $p < 0.000001$
- Statistically significant positive relationship

Linear Regression:

- Coefficient = +4,495 visits per unit PM2.5
- $p < 2 \times 10^{-16}$

Model Selection

Three models were evaluated on the merged dataset:

Model	Variables	R^2 / Var. Explained
Linear Regression	PM2.5 only	13.2%
K-Means Clustering	PM2.5, Visits	N/A
Random Forest (RF1)	PM2.5 only	74.45%
Random Forest (RF2)	All 9 features	76.99%

Why Random Forest?

- Captures non-linear threshold effects in PM2.5
- Handles correlated predictors gracefully
- Provides variable importance ranking

What Drives Healthcare Visits?

Random Forest Variable Importance (%IncMSE —drop in accuracy when variable is removed):

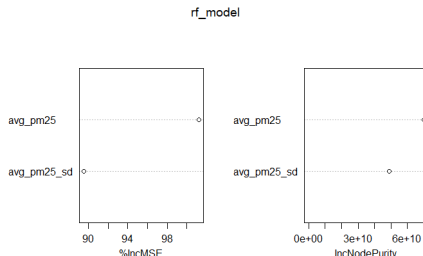


Figure 3: Variable importance from Random Forest (rf_model3, $n = 500$ trees)

Top predictors: avg_pm25 > avg_pm25_sd > pct_mychart

Convergent evidence across all methods:

Method	Finding	Strength
Correlation	$r = 0.29, p < 0.001$	Significant
Linear Regression	+4,495 visits per PM2.5 unit	$p < 2 \times 10^{-16}$
Random Forest	PM2.5 = (%IncMSE $\approx$ 35%)	77%
K-Means	3 clusters separated by PM2.5 threshold	~ 7.5 PM2.5

Achievements:

- Successfully linked air pollution to healthcare Hospital Visits across Kansas census tracts
- Random Forest (77% variance explained) outperforms linear approaches
- PM2.5 is the strongest single predictor —even after controlling for race, smoking, and demographics
- Identified a non-linear threshold effect at $PM_{2.5} \approx 7.5$

The optimal model used $mtry = 5$, selected by minimum RMSE.

Final cross-validated performance: RMSE = 2,779.4,
MAE = 1,233.8, $R^2 = 0.8039$.

$mtry$	RMSE	R^2	MAE
2	2,786.6	0.8036	1,241.9
5	2,779.4	0.8039	1,233.8
9	2,843.3	0.7953	1,261.2

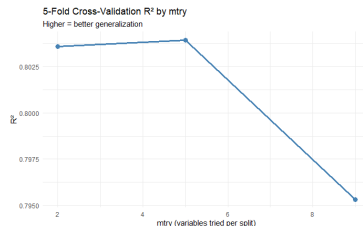


Figure 4: Cross Validation

Why This Matters

- **PM2.5 is not purely an environmental issue.** It directly affects **healthcare demand**.
- Higher pollution levels are associated with **more hospital visits**.
- In our analysis, PM2.5 emerged as the **strongest predictor**.
- We also identified a **non-linear threshold** around **$7.5 \mu\text{g}/\text{m}^3$** .

Key question: How can hospitals use this information in practice?

Targeted Resource Allocation

- Hospitals can then:
 - adjust **staffing levels**,
 - pre-position **respiratory support** such as inhalers and oxygen therapy,
 - strengthen **local care capacity**.
- This supports a **spatially targeted allocation of resources**.

Prevention Opportunities

- These findings also open the door to **preventive strategies**.
- Vulnerable populations could be **alerted in advance**.
- Preventive recommendations may help reduce **avoidable hospital visits**.
- In this way, environmental data can support both **preparedness** and **prevention**.

Main Message

PM2.5 is not only statistically significant, but also **actionable**. It can help hospitals anticipate demand, allocate resources more efficiently, and support preventive interventions in the areas that need them most.

This opens the door to proactive prevention strategies and targeted resource allocation.

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Introduction

Five particulate matter (PM_{2.5}) is one of the most well-documented environmental threats to **public health**, linked to respiratory illness, cardiovascular disease, and increased mortality. While the health effects of **air pollution** are locally established, the extent to which geographic variation in PM_{2.5} translates into measurable differences in healthcare utilization within a single state remains understudied. Kansas presents a compelling case: its census tracts vary substantially in pollution exposure, demographic composition, and access to digital health tools. This study integrates tract-level PM_{2.5} estimates with hospital visit records to quantify how **air quality shapes healthcare demand** across the state.

Objective

This study aimed to quantify the association between **average PM_{2.5} concentration** and average healthcare visits per person at the census-tract level across Kansas, while identifying the sociodemographic factors that moderate or amplify this relationship. A secondary goal was to determine whether a non-linear machine learning approach outperforms traditional linear regression in capturing this relationship, and ultimately to provide a data-driven framework for the proactive, spatially targeted allocation of healthcare resources.

Methodology

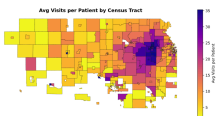
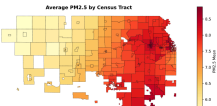
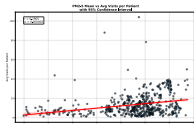
Data Collection: Two datasets were used: Kansas PM_{2.5} pollution predictions (2016-2020) aggregated to census-tract means, and hospital visit counts by census block group, enriched with patient demographics and social determinants.

Data Cleaning: Census tracts with more than 30,000 total visits were removed as outliers to ensure model stability and avoid skewed predictions.

Data Linkage: The 12-digit census block code was trimmed to 11 digits to align with the tract-level pollution identifier, enabling a clean merge of both datasets into 1,248 observations.

Feature Engineering: Five tract-level demographic proportion variables were constructed, capturing race, ethnicity, smoking status, marital status, and MyChart digital health activation rate.

Model Evaluation: Four models were tested: linear regression, K-Means clustering, and two Random Forest variants (PM_{2.5} only vs. all 8 featured). The full 8-feature Random Forest model was selected as the best performer based on variance explained.



Results

The cross-validated Random Forest explained 80.6% of the variance in total healthcare visits ($R^2 = 0.8039$), with an RMSE of 2,775 and MAE of 1,214 – a dramatic improvement over linear regression's R^2 of 13.2%. PM_{2.5} emerged as the single strongest predictor by variable importance, followed by MyChart activation rate, average screenings, and Hispanic population proportion. Partial dependence analysis revealed a non-linear threshold: visit counts accelerate sharply above PM_{2.5} $\approx 7.5 \mu\text{g}/\text{m}^3$, explaining why linear regression failed to capture the relationship. This threshold was independently confirmed by K-Means clustering. Statistically, PM_{2.5} showed a significant positive correlation with visits ($r = 0.29$, $p = 0.000001$), and a linear regression coefficient of $+4.455$ visits per unit PM_{2.5} ($p = 2 \times 10^{-16}$). Choropleth maps further revealed a clear geographic overlap between high-PM_{2.5} zones and high visit burden, concentrated in eastern Kansas.

Analysis

Evidence converged consistently across all methods – correlation, regression, Random Forest, spatial mapping, and clustering – pointing to PM_{2.5} as the dominant driver of healthcare utilization in Kansas. The non-linear threshold at PM_{2.5} $\approx 7.5 \mu\text{g}/\text{m}^3$ is operationally significant: below it, visit burden remains relatively stable, above it, demand rises sharply, placing acute strain on local healthcare infrastructure. This threshold effect explains why linear regression ($R^2 = 13.2\%$) failed to capture the relationship that the Random Forest ($R^2 = 80.6\%$) successfully modeled, validating the use of a non-linear ensemble approach for this type of health-environment data.

$$C_{\text{PM}_{2.5}} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2} \quad \hat{\beta} = \frac{1}{B} \sum_{b=1}^B f_b(x)$$

Conclusion

This study shows that PM_{2.5} is a key driver of healthcare demand across Kansas, with a non-linear increase in visits above critical pollution levels. These findings enable proactive, spatially targeted resource allocation, allowing hospitals to anticipate demand, optimize staffing, and support high-risk areas. Additionally, PM_{2.5} can serve as an early warning signal to support preventive interventions and reduce avoidable hospital visits. High-risk tracts—characterized by elevated PM_{2.5} and limited digital health access—should be prioritized for targeted interventions. Overall, integrating environmental data into healthcare planning supports a shift toward more efficient and equitable healthcare systems.

Data

Source: US Census Bureau and Environmental Protection Agency (EPA) 2016-2020 Kansas PM_{2.5} Predictions, 2016-2020
 https://www.kansas.gov/health-and-welfare/health-data/kansas-pm25-predictions-2016-2020

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x})$$

$$\mathcal{L}_{\text{split}} = \min_{j, t} \left[\sum_{i \in R_1(j, t)} (y_i - \bar{y}_{R_1})^2 + \sum_{i \in R_2(j, t)} (y_i - \bar{y}_{R_2})^2 \right]$$

- $\hat{y}$ —predicted number of total healthcare visits
- B —number of trees in the forest ($B = 500$)
- $T_b(\mathbf{x})$ —prediction from the b -th decision tree
- $\mathbf{x}$ —feature vector:
(avg_pm25, pct_mychart, pct_hispanic, ...)